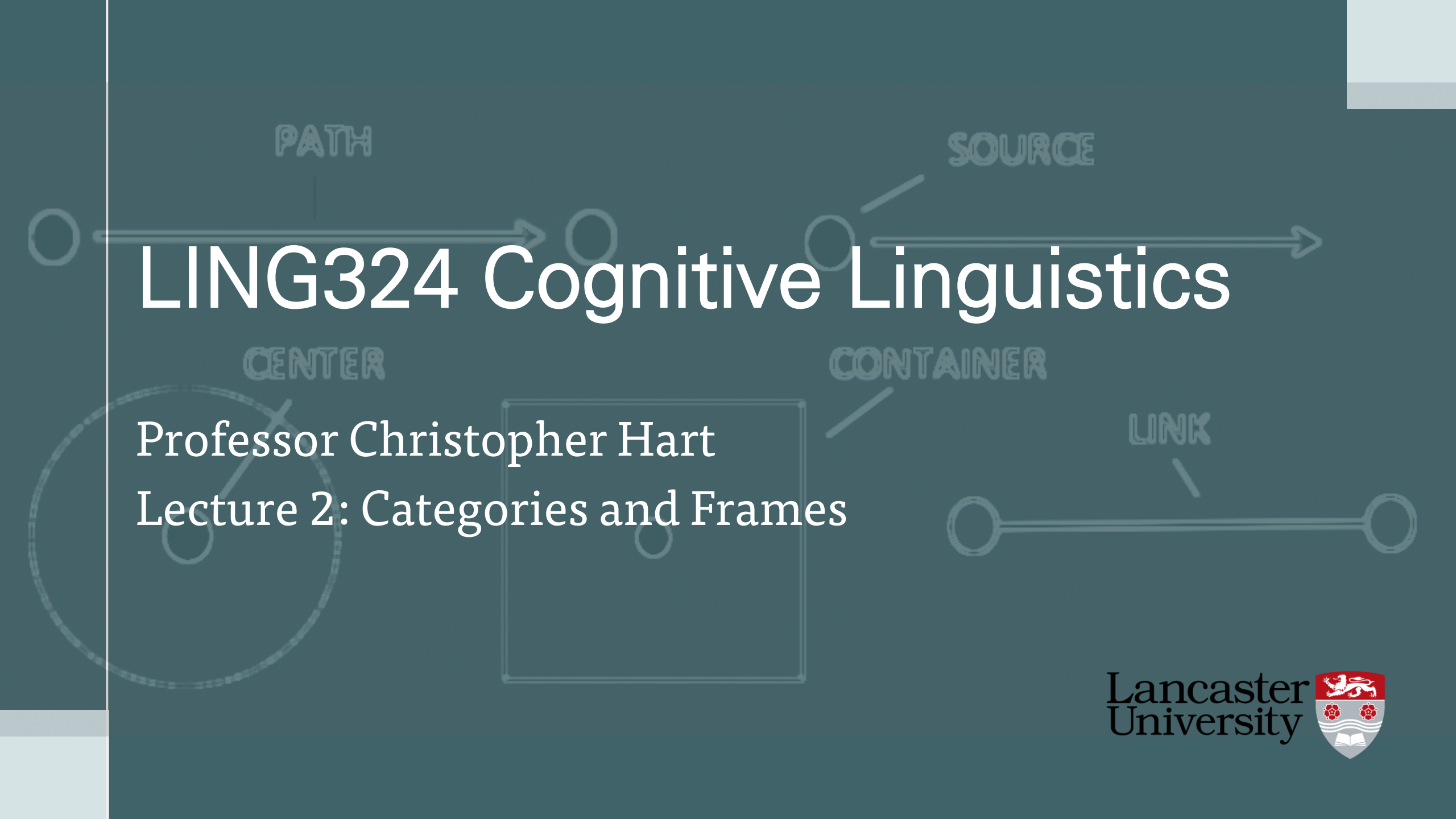




LING324 Cognitive Linguistics

Professor Christopher Hart
Lecture 2: Categories and Frames

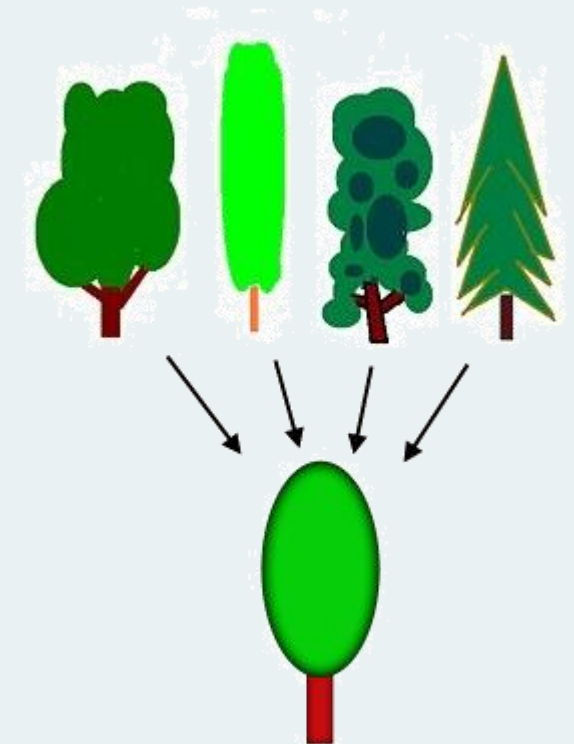
Overview

- Introduction
- Categories
 - The classic view
 - Prototypes
 - Theory of radial categories
- Frames
 - The dictionary view
 - Frames
 - Frames and categories
 - FrameNet

Introduction

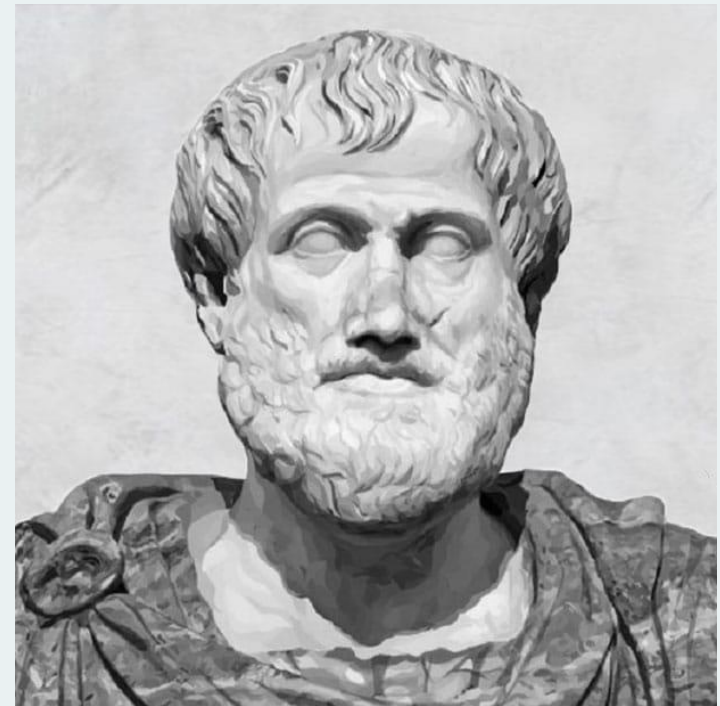
The act of categorisation is one of the most basic human cognitive activities. Categorisation involves the apprehension of some individual entity, some particular of experience, as an instance of something conceived more abstractly that also encompasses other actual and potential instantiations. (Croft & Cruse 2004: 74)

- Categorisation a central topic in philosophy, psychology and linguistics
- Cognitive tool implicated in learning, planning, communication and economy of knowledge (Croft & Cruse 2004: 74)



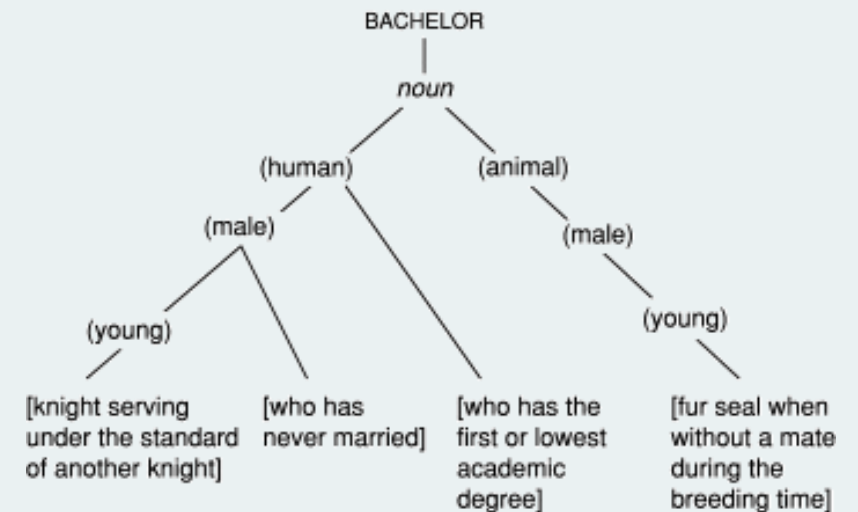
The Classic View

- The 'classical model' traceable to Aristotle
- Categories defined by sets of **necessary and sufficient** features
 - Necessary in so far as required of all members
 - Sufficient in so far as enough to guarantee membership
- Found in formal and structuralist approaches to lexical semantics (Geeraerts 2010)



The Classic View

- E.g., BACHELOR = [ADULT +] [MARRIED -] [MALE +]
- From this perspective:
 - Categories are binary
 - Boundaries between categories are clearly defined
 - All members of a category are equal representatives

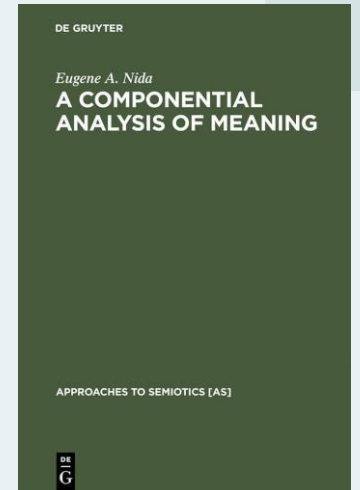


The Classic View

- Decompositional semantics or **componential** analysis of meaning

	HUMAN	ADULT	FEMALE
<i>Man</i>	+	+	-
<i>Woman</i>	+	+	+
<i>Boy</i>	+	-	-
<i>Girl</i>	+	-	+

A meaning is not a thing in itself, but only a set of contrastive relations. There is no way to determine a meaning apart from comparisons and contrasts with other meanings within the same semantic set. (Nida 1975: 151)



The Classic View

game

The Classic View

- Problems with the classic view: E.g. GAME
 - **Definition** – not always possible to arrive at set of necessary and sufficient features
 - **Fuzziness** – boundaries between some categories blurred
 - **Prototypicality** – some members judged as 'better' or more representative examples than others

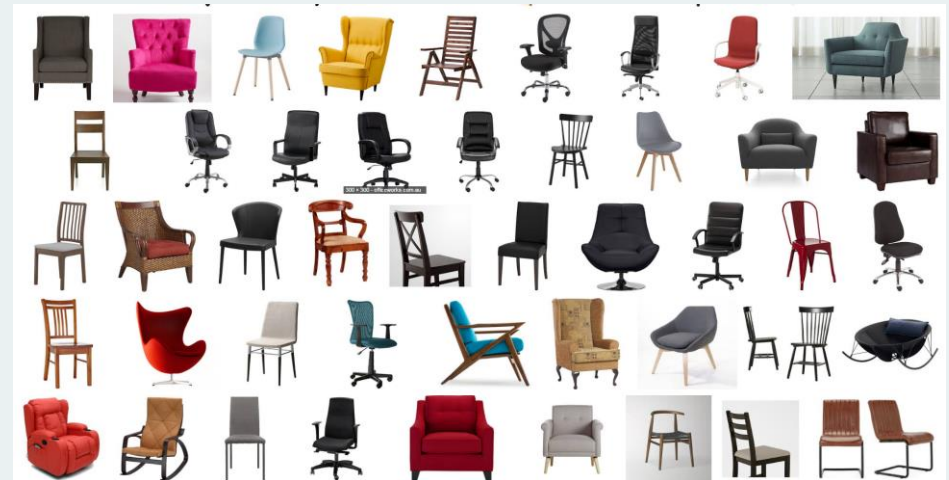


The Classic View

- Problems with the classical model: E.g. CHAIR

(Object), (Physical), (Non-living), (Artifact),
(Furniture), (Portable), (Something with legs),
(Something with a back), (Something with a seat),
(Seat for one)

(Katz 1972)



Prototypes

- Classical model debunked by experimental research of Eleanor Rosch (e.g. Rosch 1973, 1978; Rosch & Mervis 1975)
- Developed alternative **prototype** model of category structure
- **Goodness-Of-Exemplar (GOE)** experiments
 - Subjects presented with a category and a list of putative members which asked to rate, on a 1-7 scale, according to how good an example they are of the category
 - Combining results across subjects shows graded structure organised around a prototype

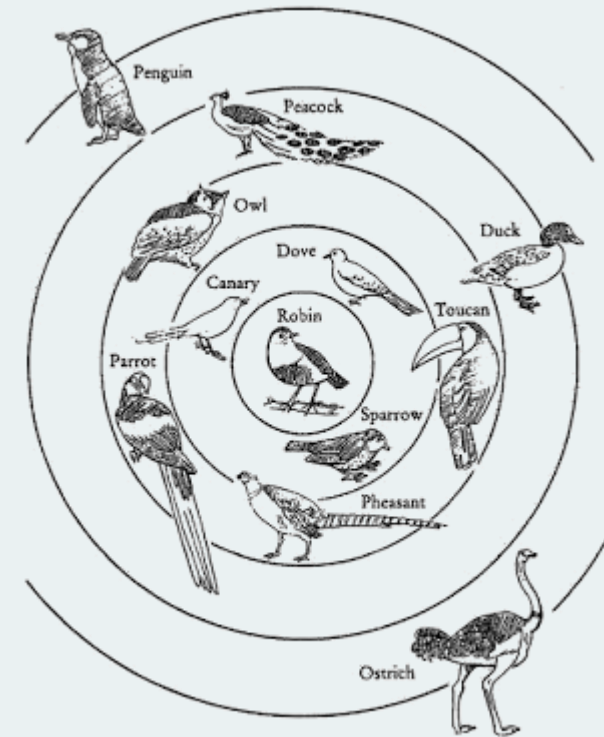


Figure 1 Birdiness rankings

Prototypes

Rank	BIRD	FRUIT	FURNITURE
Top five (from more to less representative)			
1	Robin	Orange	Chair
2	Sparrow	Apple	Sofa
3	Bluejay	Banana	Couch
4	Blue bird	Peach	Table
5	Canary	Pear	Easy chair
Bottom five (from more to less representative)			
5	Turkey	Nut	Closet
4	Ostrich	Gourd	Vase
3	Emu	Olive	Ashtray
2	Penguin	Pickle	Fan
1	Bat	Squash	Telephone

(Rosch 1975: appendix)

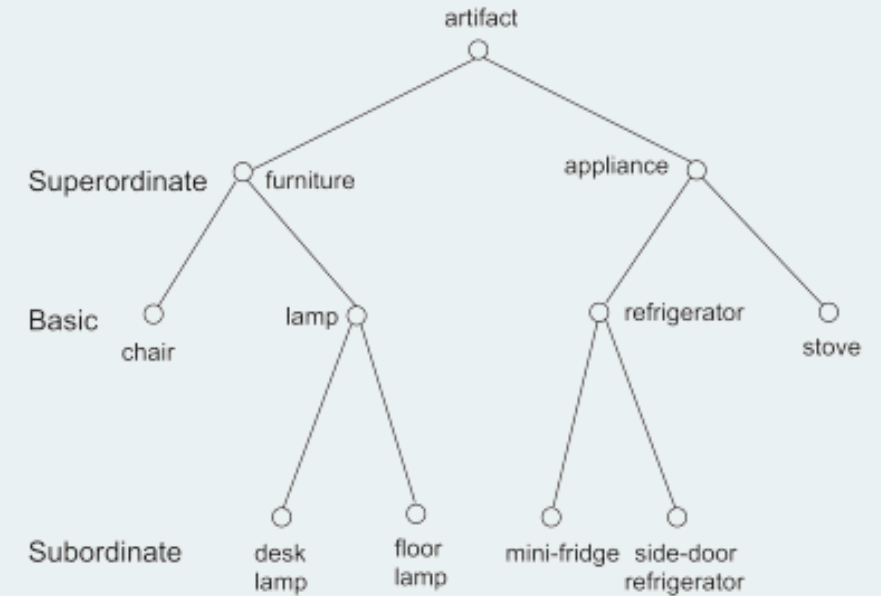
Prototypes

- GOE ratings culture dependent
- Correlate with independent properties:
 - Frequency and order of mention
 - Order of learning
 - Family resemblance
 - Verification speed
 - Priming



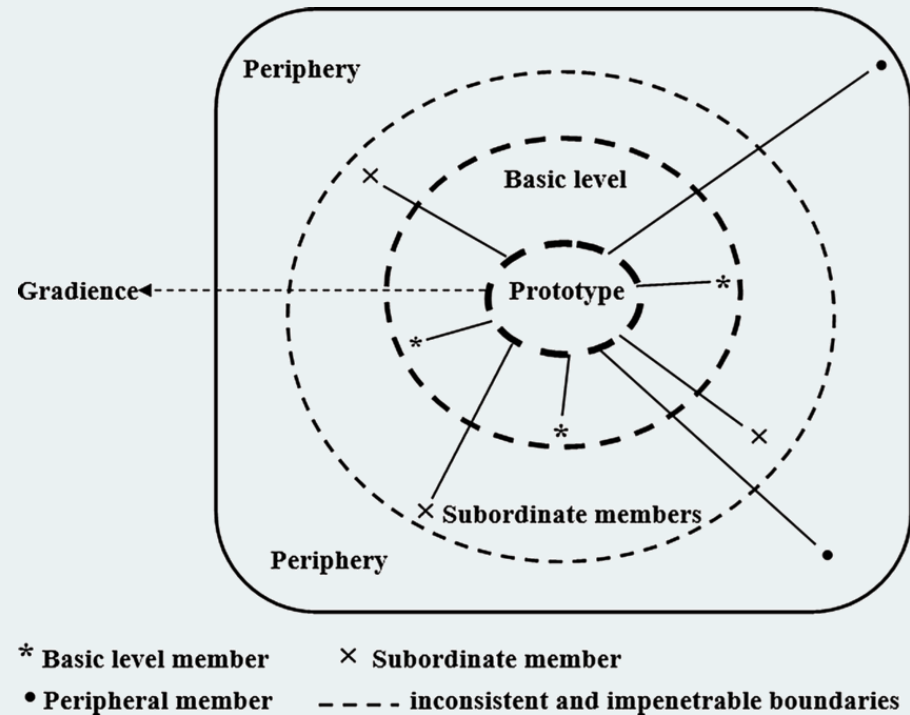
Prototypes

- Levels of categorisation – more specific categories nested in more inclusive ones
 - E.g. fruit > **apple** > granny smith
- **Basic level** categories defined by substantive characteristics:
 - Level at which patterns of behavioural interaction most identifiable
 - Level at which clear visual image can be formed
 - Level at which part-whole information represented
 - Level used for everyday reference
 - Level to which subordinate items most rapidly categorised as belonging



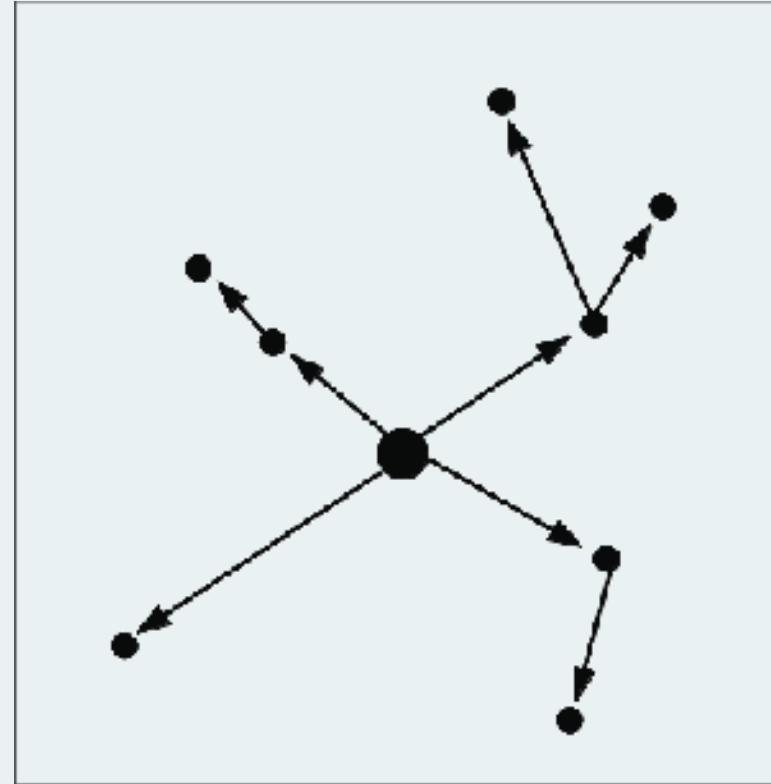
Theory of Radial Categories

- Model developed in CL as explicit theory of knowledge representation accounting for prototype effects (Lakoff 1987; Taylor 2003)
- Categories are conceptual structures in which a range of lexicalised concepts are organised radially in relation to a central prototype
- Gradient structure and fuzzy boundaries



Theory of Radial Categories

- Members connected to prototype directly or via series of intermediaries
- Extensions outward motivated
- Distant members may lack obvious connection
- Relations between words and between senses of same word – polysemy



Theory of Radial Categories

- Prototypicality observed at all levels of linguistic structure “from phonology to morphology to syntax to the lexicon” (Lakoff 1987: 67)
- Applies across ontological and lexical categories:
 - THINGS (nouns)
 - ACTIONS (verbs)
 - RELATIONS (prepositions)



Radial structure of the polysemous verb *drop*
(Halas 2016: 129)

Theory of Radial Categories

- Category structure and prototype effects born of **idealised cognitive models** (cf. frames) – conceptual structures by means of which we organise our knowledge (Lakoff 1987: 68)
- Abstractions across experience and thus degrees of 'fit' with potential instantiations
- Prototypicality arises where more or less perfect fit between ICM and specific case
 - E.g. BACHELOR < MARRIAGE
 - The Pope / Tarzan

Theory of Radial Categories

- In many cases, prototypicality dependent on combinations of ICMs – **cluster models** (Lakoff 1987: 74-76)
- E.g. MOTHER
 - BIRTH MODEL / GENETIC MODEL / NURTURANCE MODEL / MARITAL MODEL / GENEALOGICAL MODEL
 - *Stepmothers, surrogate mothers, adoptive mothers, biological mothers* etc. all mothers but less prototypical by virtue of deviating from one or more ICM

The 'Dictionary' View

- Linguistic meaning, as represented in the lexicon, is distinct from world knowledge
- Semantics is concerned with 'core' linguistic meaning only - **denotation**
- **Connotation** not included
- The lexicon is characterised as a kind of mental dictionary
- Entries defined in terms of semantic features or components

A	
¹ a \ˈā\ <i>n</i> , <i>pl</i> a's or as \ˈāz\ : 1st letter of the alphabet	abate-ment \əˈbātment\ <i>n</i> : tax reduction
² a \ə, ˈā\ <i>indefinite article</i> : one or some — used to indicate an unspecified or unidentified individual	ab-at-toir \ˈabə,twār\ <i>n</i> : slaughterhouse
aard-vark \ˈärd,värk\ <i>n</i> : ant-eating African mammal	ab-bess \ˈabəs\ <i>n</i> : head of a convent
aback \əˈbak\ <i>adv</i> : by surprise	ab-bey \ˈabē\ <i>n</i> , <i>pl</i> -beys : monastery or convent
aba-cus \ˈabəkəs\ <i>n</i> , <i>pl</i> aba-ci \ˈabəsī, -kē\ or aba-cus-es : calculating instrument using rows of beads	ab-bot \ˈabət\ <i>n</i> : head of a monastery
abaft \əˈbaft\ <i>adv</i> : toward or at the stern	ab-bre-vi-ate \əˈbrēvē,āt\ <i>vb</i> -at-ed ; -at-ing : shorten — ab-bre-vi-a-tion \əˈbrēvēˈāshən\ <i>n</i>
ab-a-lo-ne \ˈabəˈlōnē\ <i>n</i> : large edible shellfish	ab-di-cate \ˈabdi,kāt\ <i>vb</i> -cat-ed ; -cat-ing : renounce — ab-di-ca-tion \ˈabdiˈkāshən\ <i>n</i>
¹ aban-don \əˈbandən\ <i>vb</i> : give up without intent to reclaim — aban-don-ment <i>n</i>	ab-do-men \ˈabdəmən, abˈdömən\ <i>n</i> 1 : body area between chest and pelvis 2 : hindmost part of an insect — ab-dom-i-nal \ˈabˈdämənˈl\ <i>adj</i> — ab-dom-i-nal-ly <i>adv</i>
² abandon <i>n</i> : thorough yielding to impulses	ab-duct \ˈabˈdʌkt\ <i>vb</i> : kidnap — ab-duc-tion \-ˈdʌkshən\ <i>n</i> — ab-duc-tor \-tər\ <i>n</i>
aban-doned \əˈbandənd\ <i>adj</i> : morally unrestrained	abed \əˈbed\ <i>adv</i> or <i>adj</i> : in bed
abase \əˈbās\ <i>vb</i> abased ; abas-ing : lower in dignity — abase-ment <i>n</i>	ab-er-ra-tion \ˈabəˈrāshən\ <i>n</i> : deviation or distortion — ab-er-rant \ˈabərənt\ <i>adj</i>
abash \əˈbāsh\ <i>vb</i> : embarrass — abashment <i>n</i>	abet \əˈbet\ <i>vb</i> -tt- : incite or encourage — abet-tor , abet-ter \-ər\ <i>n</i>
abate \əˈbāt\ <i>vb</i> abat-ed ; abat-ing : decrease or lessen	

The 'Dictionary' View

discount

Frames

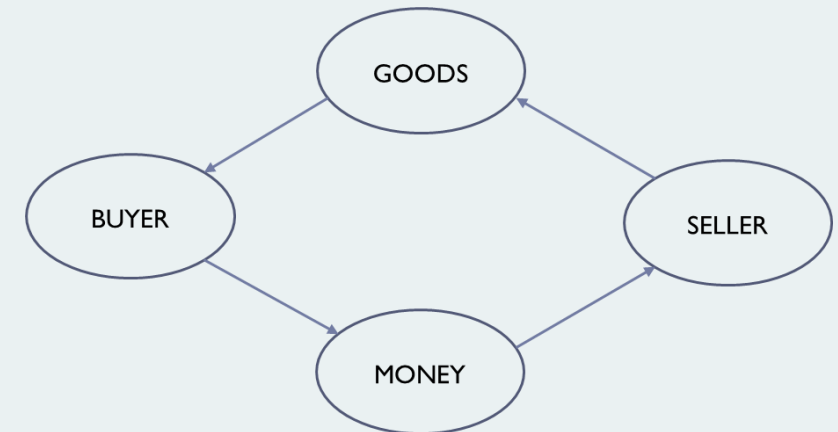
- Definition possible only with reference to specific context – **frame**
- Structured bundles of knowledge representing real-world situations, arrangements, scenarios etc.
- Encyclopaedic
- Inclusive of function, affect etc.
- Open-ended but accessed with different degrees of salience

The particular organisation of knowledge which stands as a prerequisite to our ability to understand the meaning of ... associated words (Fillmore 1985: 224)

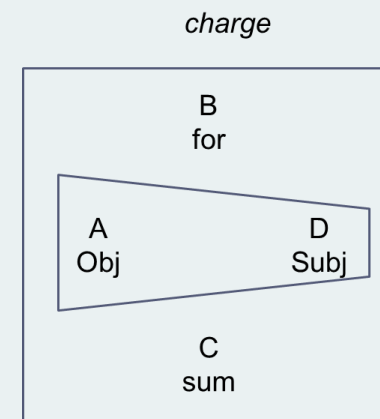
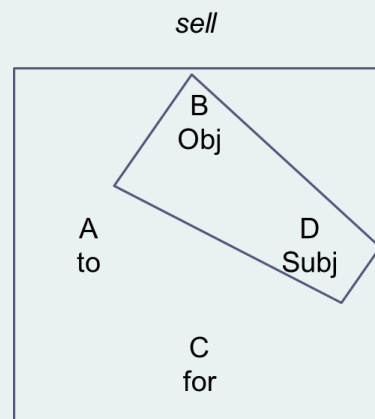
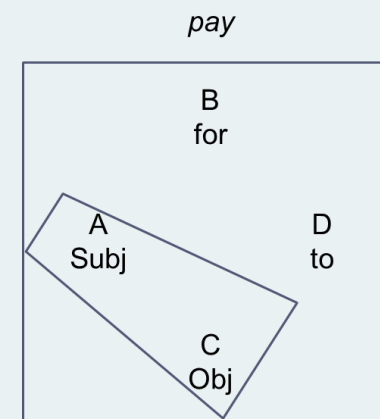
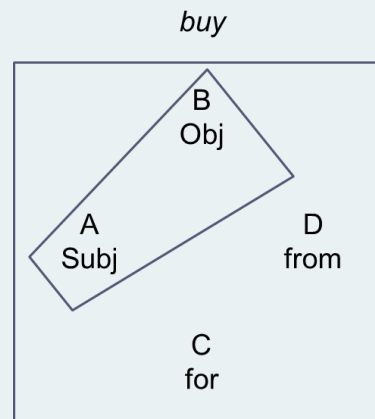
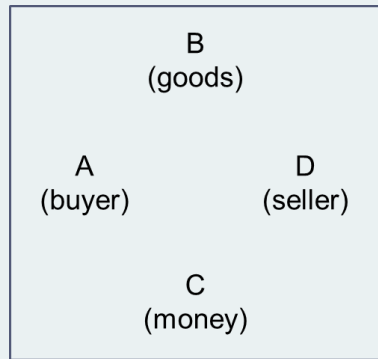
Any system of concepts related in such a way that to understand any one of them you have to understand the whole structure in which it fits (Fillmore 1982: 111)

Frames

- COMMERCIAL EVENT frame
 - Words *buy, sell, pay, spend, cost, charge, tender, change* and *discount* meaningful only within COMMERCIAL EVENT frame
 - Minimally made up of four **core elements**: BUYER, SELLER, MONEY, GOODS
 - Words foreground roles and relations within frame and encode perspective
 - Accounts for argument structure (**valence**)



Frames



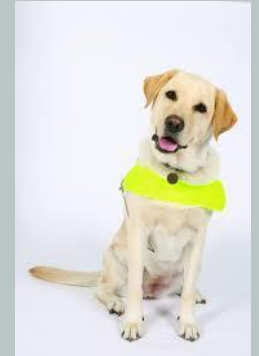
Frames

- Frames provide perspective
 - E.g. *coast* vs. *shore*
 - LAND frame vs. SEA frame
- Lexical choice imposes a particular **framing**
 - E.g. *stingy* vs. *thrifty*



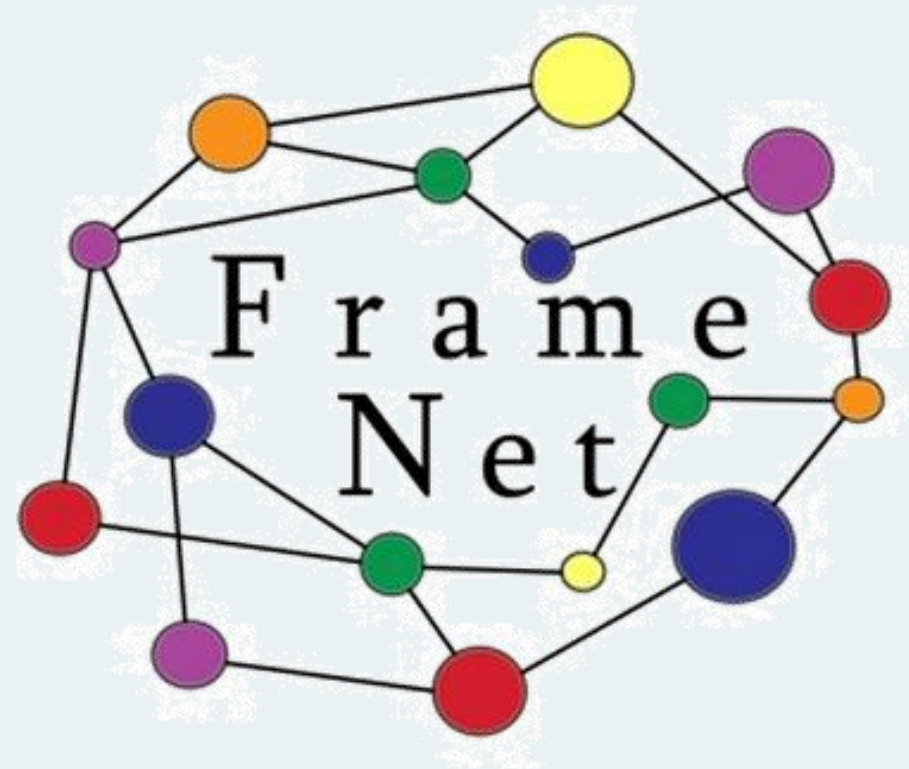
Frames and Categories

- Frames disambiguate polysemous words
 - E.g. *menu*
 - Meaning dependent on currently active frame
- Frames represent prototypical scenarios
 - E.g. MARRIAGE CEREMONY or RESTAURANT
- Prototypes provide **default assignments** for frame elements
 - Not fixed but dependent on currently active frame
 - E.g. *ball* or *dog*



FrameNet

- Inventory of frames, their elements and interrelations
 - **Core FEs:** elements essential to the meaning of a frame
 - **Non-core FEs:** elements providing more circumstantial information
 - **Frame-Frame relations:** inheritance, sub-frame, metaphor



FrameNet



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FrameNet Data

[Appeal](#)
[Appellations](#)
[Apply_heat](#)
[Appointing](#)
[Architectural_part](#)
[Arithmetic](#)
[Arithmetic_commutative](#)
[Arithmetic_non-commutative](#)
[Armor](#)
[Arraignment](#)
[Arranging](#)
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[Arriving](#)
[Arson](#)
[Artifact](#)
[Artifact_subpart](#)
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[Artistic_style](#)
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[Asymmetric_reciprocity](#)
[Atonement](#)
[Attaching](#)
[Attack](#)
[Attempt](#)
[Attempt_action_scenario](#)
[Attempt_distant_interaction_scenario](#)
[Attempt_means](#)
[Attempt_obtain_food_scenario](#)
[Attempt_obtain_mineral_scenario](#)

Arrest

[Lexical Unit Index](#)

Definition:

Authorities charge a **Suspect**, who is under suspicion of having committed a crime (the **Charges**), and take him/her into custody.
The police **ARRESTED** Harry on charges of manslaughter.

FEs:

Core:

Authorities [Auth] The **Authorities** charge the **Suspect** with committing a crime, and take him/her into custody.
Semantic Type: Sentient The police **ARRESTED** Harry on charges of manslaughter.

Charges [Chrg] **Charges** identifies a category within the legal system; it is the crime with which the **Suspect** is charged.
The police **ARRESTED** Harry on charges of manslaughter.

Offense [Off] **Offense** identifies the ordinary language use of the reason for which a **Suspect** is arrested.
They arrested Harry for shoplifting.

Suspect [Susp] The **Suspect** is taken into custody, under suspicion of having committed a crime.
The police **ARRESTED** Harry on charges of manslaughter.

Thank you

Selected References

- Barsalou, L.W. (1985). Ideals, central tendency, and frequency of instantiation as determinants of graded structure in categories. *Journal of Experimental Psychology* 11 (4): 629-54.
- Fillmore, C. (1975). An alternative to checklist theories of meaning. *Proceedings of the First Annual Meeting of the Berkely Linguistics Society*. North Holland. pp. 123-31.
- Fillmore, C. (1977). Scenes and frame-semantics. In A. Zampoli (ed.), *Linguistic Structures Processing*. North Holland. pp. 55-82.
- Fillmore, C. (1982). Frame semantics. In Linguistics Society of Korea (ed.), *Linguistics in the Morning Calm*. Seoul: Hanshin Publishing. pp.111-37. Reprinted in D. Geeraerts (ed.) (2006), *Cognitive Linguistics: Basic Readings*. Mouton. pp. 373-400.
- Fillmore, C. (1985). Frames in the semantics of understanding. *Quaderni di Semantica* 6: 222-54.
- Lakoff, G. (1987). *Women, Fire and Dangerous Things: What Categories Reveal About the Mind*. University of Chicago Press.
- Rosch, E. (1975). Cognitive representations of semantic categories. *Journal of Experimental of Psychology* 104 (3): 192-233.
- Rosch, E. (1977). Human categorization. In N. Warren (ed.), *Studies in Cross-linguistic Psychology*. Academic Press. pp. 1-49.
- Rosch, E. (1978). Principles of categorization. In B. Loyd and E. Rosch (eds.), *Cognition and Categorization*. Hillsdale, NJ: Erlbaum. pp. 27-48. Reprinted in E. Margolis and S. Laurence (eds.) (1999), *Concepts: Core Readings*. MIT Press. pp. 189-206.
- Rosch, E. and Mervis, C. (1975). Family resemblances: Studies in the internal structure of categories. *Cognitive Psychology* 7: 573-605.
- Rosch, E., Mervis, C., Gray, W., Johnson, D. and Boyes-Braem, P. (1976). Basic objects in natural categories. *Cognitive Psychology* 8: 382-439.
- Taylor, J. (2003). *Linguistic Categorization*, 3rd edn. Oxford University Press.